

Technical Report for Road Extraction ArcGis Integration

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1 Preamble

This document provides an overview of the results obtained from the road extraction project. Its primary purpose is to present the technical findings of the developed model, including the rationale behind the selected architecture, model performance and evaluation results. A separate technical guide is also provided, offering detailed instructions on how to deploy and use the tool within ArcGIS.

2 Introduction

Roads, whether paved, gravel, or intended for logging, are a fundamental component of modern society. Road infrastructure has a profound impact on economic development and competitiveness by enabling economic growth. They facilitate the transport of goods and people across vast regions, supporting complex economic and social structures. Logging roads, consisting of dirt or gravel surfaces, are used by logging companies to access sites and transport materials. However, roads also profoundly affect local ecosystems by altering habitats, increasing animal mortality through collisions, and disrupting migration and mating patterns. Runoff from road salting and large-scale land clearing can further degrade surrounding flora and fauna. Compacted soil caused by machinery, such as that used during logging operations, has a negative impact both economically and environmentally. The negative impacts of these roads can be mitigated through improved management and monitoring, which remains a complex task due to the involvement of multiple organizations. On-site visits to these roads are useful for various purposes, such as analyzing their environmental and economic impacts. These visits are also necessary to confirm road usability and current condition in order to assess their overall impact. This requires detailed maps of road networks to enable effective tracking and monitoring, which is challenging due to the quantity and diversity of roads. While paved roads (e.g., highways) are generally well documented by government agencies, many unmapped or poorly maintained roads, particularly logging roads, remain untracked. Since these roads are often privately owned and distributed across different corporations and organizations, aggregating their locations into a centralized map is challenging. Remote sensing provides a promising approach for large-scale road surveying without the need to aggregate and verify data from multiple sources. Remote sensing techniques, such as satellite and drone imagery, enable coverage of large areas quickly and efficiently without requiring on-site visits. However, due to the vast extent of logging and secondary roads, manually analyzing and mapping roads from remote sensing data remains challenging. Manual tracing and extraction can be accurate but are highly time-consuming, as they require tedious annotation across thousands of images. New-Brunswick, covering nearly 73,000 km², presents additional challenges due to the abundance of logging roads and continuously changing environmental conditions. This project explores the use of automated techniques to extract roads from remote sensing data in New-Brunswick, Canada. Artificial Intelligence, particularly Deep Learning (DL), has already demonstrated strong

performance in similar tasks. DL-based segmentation enables the extraction of objects from images, such as roads, with a high degree of accuracy and reliability. This project was divided into three phases: an exploratory phase, a large-scale study of DL algorithms, and a final fine-tuning and integration phase. The exploratory phase consisted of initial model training to assess the feasibility of the approach and yielded promising results. The second phase involved a large-scale study of DL using various algorithms and techniques on a larger annotated dataset. The results were highly promising and were used to generate the final dataset for fine-tuning and integration. Previous technical reports provided an overview of the exploratory and large-scale phases, including key results and performance. However, the focus of this report is placed on the final model, its performance, and the fine-tuning process used for integration. The outline of this technical report is presented below.

Report Outline

1. Model Creation

This section presents the dataset, preprocessing pipeline, model architecture, training procedure and evaluation methodology used to develop the road segmentation model.

2. Model Performance

This section evaluates the segmentation performance of the final model using metrics such as Accuracy, Precision, Recall, F1-score, IoU and PPA. Best, average and worst-performing cases are also analyzed to better understand model behaviour.

3. Threshold Sensitivity Analysis

This section investigates the impact of different probability thresholds on segmentation performance. Precision, Recall, F1-score, IoU, false positive rates and false negative rates are analyzed to determine an appropriate deployment threshold.

4. Model Efficiency and Deployment Performance

This section evaluates the computational efficiency of the proposed solution, including ONNX inference speed, tile processing times, mosaicking performance and large-scale deployment estimates for CSRNO and New-Brunswick.

5. Explainability

This section presents Grad-CAM explainability analyses used to visualize the image regions contributing most strongly to road predictions. These results provide insight into model decision-making and help validate that the model focuses on meaningful road structures.

3 Model Creation

This section presents the proposed model configuration, including the dataset, preprocessing pipeline and training parameters used for road segmentation.

3.1 Data

DL-based segmentation models require annotated datasets, often consisting of thousands of samples, in order to achieve strong generalization performance. Earlier experiments and related work demonstrated that DL-based segmentation can effectively extract logging roads from high-resolution satellite imagery. As such, CCNB-Innov annotated large areas of interest containing logging roads, paved roads and other transportation pathways. These samples were manually annotated and verified, increasing the reliability and validity of the dataset. The selected regions were collected from the CSRNO area and primarily consisted of agricultural, forested and rural environments.

In total, 39 regions were selected based on various criteria and manually annotated. Each case covered approximately 196 km^2 , resulting in a total annotated area of approximately 7644 km^2 . The imagery was obtained using the World Imagery layer within ArcGIS, which combines imagery from multiple satellite and aerial imagery databases. Using a cell size of 2, the imagery corresponded to a spatial resolution of approximately 2 meters per pixel and primarily included data from the Maxar WorldView-02 satellite. Figure 1 presents an example of the imagery used for annotation within ArcGIS.



Figure 1: Example of high-resolution imagery used for road annotation in ArcGIS.

Once annotated, the data were uploaded and combined within ArcGIS to generate corre-

sponding segmentation masks for each selected region. The imagery and masks were then exported using a cell size of 2 and an image size of 1024×1024 pixels. This corresponds to an approximate spatial coverage of $2.048 \text{ km} \times 2.048 \text{ km}$ per image tile, enabling large-area coverage while maintaining relatively high spatial resolution. The export process consisted of extracting 1024×1024 image tiles using horizontal and vertical strides of 512 pixels, enabling overlap between neighbouring tiles and increasing data diversity. In total, 2,592 matching image and annotation pairs were exported. The dataset was subsequently divided using an 80:20 split for training and validation.

3.2 Model Training

Various segmentation architectures are available with different levels of complexity and numbers of parameters. Related work has demonstrated that transformer-based and larger vision architectures can sometimes provide slightly higher segmentation performance. However, these approaches often require substantially higher computational resources and significantly larger model sizes. In this application, pixel-level boundary precision is not the primary objective. Instead, the main goal is to efficiently detect and extract the maximum number of roads over large geographic regions. As such, model selection must balance segmentation performance, computational efficiency and deployment practicality.

Based on these considerations, a U-Net architecture with a ResNet101 encoder was selected. This architecture provided strong segmentation performance while maintaining a relatively lightweight and practical model size compared to more computationally demanding architectures. The selected model can also be trained or fine-tuned locally on consumer-grade hardware, enabling rapid experimentation and optimization without requiring specialized infrastructure. The model was trained using the AdamW optimizer for 30 epochs with a learning rate and weight decay of 1×10^{-4} . The input consisted of 1024×1024 RGB image patches normalized using ImageNet statistics. Input pixel values were first scaled between 0 and 1, followed by normalization using a mean of $[0.485, 0.456, 0.406]$ and a standard deviation of $[0.229, 0.224, 0.225]$.

This normalization strategy helped reduce variations caused by differences in imagery quality, illumination conditions and sensor characteristics while improving overall training stability across heterogeneous data sources. The selected image size also enabled the model to capture broader contextual information surrounding road networks while remaining feasible to train and deploy on consumer-grade GPUs. Several data augmentation techniques were implemented to improve model generalization and robustness. These augmentations included horizontal and vertical flipping, random rotations, affine transformations, brightness and contrast adjustments, Gaussian noise injection and motion blur augmentations.

4 Results

This section evaluates the performance of the proposed road segmentation model in terms of segmentation accuracy, robustness and deployment suitability. Several metrics were used to provide a comprehensive assessment of model performance, including Accuracy, Precision, Recall, F1-score, Intersection over Union (IoU), False Positive Rate (FPR), False Negative Rate (FNR) and Pixel Proximity Accuracy (PPA).

Accuracy measures the proportion of correctly classified pixels and is defined by Equation 1.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP , TN , FP and FN represent true positives, true negatives, false positives and false negatives, respectively.

Precision evaluates the reliability of road predictions and is calculated using Equation 2.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall, also referred to as Road Pixel Accuracy, measures the proportion of annotated road pixels successfully detected by the model and is defined by Equation 3.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

The F1-score combines Precision and Recall into a single metric and is defined by Equation 4.

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (4)$$

Intersection over Union (IoU) evaluates the overlap between predicted and annotated road pixels and is calculated using Equation 5.

$$IoU = \frac{TP}{TP + FP + FN} \quad (5)$$

The False Positive Rate (FPR) and False Negative Rate (FNR) quantify the proportion of incorrectly classified pixels and are defined by Equations 6 and 7, respectively.

$$FPR = \frac{FP}{FP + TN} \quad (6)$$

$$FNR = \frac{FN}{FN + TP} \quad (7)$$

In addition to conventional segmentation metrics, Pixel Proximity Accuracy (PPA) was used to evaluate prediction quality while accounting for small spatial localization errors. Road extraction is particularly sensitive to minor positional offsets because roads are narrow linear structures and even small shifts between annotations and predictions can significantly reduce strict pixel-level metrics. PPA addresses this limitation by introducing a tolerance buffer around annotated road pixels.

PPAn0 corresponds to the conventional precision metric, where a predicted road pixel is considered correct only if it exactly overlaps an annotated road pixel. For PPAn1, a prediction is considered correct if it falls within one pixel of an annotated road pixel. Similarly, PPAn2 and PPAn5 allow localization tolerances of two and five pixels, respectively. Consequently, PPA provides additional insight into whether prediction errors originate from complete road omission or from minor spatial misalignments.

The following subsections present the quantitative performance of the proposed model, followed by analyses of threshold sensitivity, computational efficiency and model explainability.

4.1 Model Performance

During training, the model achieved a best validation F1-score of 74.50%, with a validation IoU of 59.36%, Precision of 72.09%, Recall of 77.07% and an overall Accuracy of 99.16% after 30 epochs. The best-performing epoch was subsequently selected for further analyses and integrated into the developed GUI for operational testing and large-scale prediction. This model was trained prior to the correction of a small number of annotation inconsistencies related to paved roads being incorrectly labeled within the dataset. Nevertheless, the overall performance was largely unaffected, as these annotation errors represented only a small portion of the training data and were rarely encountered throughout the dataset. A final updated model will be provided in the future once the corrected annotations have been incorporated and the training process has been repeated. Although the expected impact is relatively minor, retraining with the corrected annotations should improve dataset consistency and overall model reliability.

Table 1: Average segmentation metrics across all test cases.

Metric	Value (%)
Accuracy	98.89
Precision	63.69
Recall (Road Pixel Accuracy)	66.67
F1-Score	64.98
IoU	48.51
PPA _{n0}	63.69
PPA _{n1}	68.66
PPA _{n2}	73.23
PPA _{n5}	82.46
False Positive Rate	0.60
False Negative Rate	33.33

Using all available cases, including both training and validation samples, the model achieved the average metrics presented in Table 1. Overall, the results demonstrate that the proposed model was capable of successfully extracting the majority of agricultural and forestry roads while maintaining a very low number of false detections. The average Accuracy of 98.89% indicates that the model correctly classified nearly all pixels within the evaluated images. However, because road pixels represent only a small fraction of the total image area, Accuracy alone can provide an overly optimistic assessment of segmentation performance. Consequently, metrics such as Precision, Recall, F1-score and IoU provide a more representative evaluation of road extraction quality. The average F1-score of 64.98% and IoU of 48.51% indicate a moderate-to-strong overlap between predicted and annotated roads across all test cases. Similarly, the average Recall of 66.67% shows that approximately two-thirds of all annotated road pixels were successfully detected by the model. The average Precision of 63.69% indicates that most predicted road pixels corresponded to actual roads, while maintaining a relatively low number of false detections. This behaviour is further supported by the extremely low False Positive Rate of only 0.60%, demonstrating that the model rarely classifies background regions as roads. On the other hand, the relatively high FNR of 33.33% indicates that approximately one-third of annotated road pixels were classified as background. While this may initially appear significant, it should be interpreted alongside the PPA metrics. Road extraction is particularly sensitive to small spatial offsets because roads occupy only a few pixels in width. Consequently, predictions that are visually correct may still be penalized as false negatives if they are shifted by only a small number of pixels relative to the annotations. Additionally, pixel-perfect annotations were not required during dataset creation, introducing a degree of uncertainty along road boundaries. The Pixel Proximity Accuracy (PPA) metrics provide additional insight into segmentation quality by accounting for small localization errors. While the strict pixel-level score PPA_{n0} achieved 63.69%, the value increased to 68.66%, 73.23% and 82.46% for tolerances of 1, 2

and 5 pixels, respectively. This substantial increase suggests that many apparent segmentation errors originate from small spatial offsets rather than complete road omission. The substantial increase from 63.69% PPA_{n0} to 82.46% PPA_{n5} also suggests that many of the false negatives are attributable to minor localization differences rather than complete failures to detect roads.

Such behaviour is expected in road segmentation tasks because roads represent narrow linear structures where even small prediction shifts can significantly affect conventional pixel-level metrics. Furthermore, pixel-perfect annotations were not required during dataset creation, as the primary objective was to identify road locations rather than precisely delineate every road boundary. Consequently, some disagreement between predictions and annotations may originate from annotation variability rather than true segmentation errors. The strong increase observed between PPA_{n0} and PPA_{n5} therefore indicates that the model generally identifies the correct road locations even when exact pixel alignment is not achieved. Another important observation is the relationship between Recall and the False Negative Rate. While the model successfully detected approximately 66.67% of annotated road pixels, approximately 33.33% remained undetected. These missed detections generally correspond to narrow roads, visually ambiguous roads, partially occluded roads or road segments with limited contrast relative to the surrounding environment. Nevertheless, the low False Positive Rate indicates that the model remains relatively conservative, favouring prediction reliability over excessive road generation.

Overall, the results indicate that the proposed model achieves a strong balance between detection capability and prediction reliability. The model successfully extracts most forestry and agricultural roads while maintaining a low level of false detections, making it suitable for large-scale operational road mapping applications. The PPA results further suggest that a considerable portion of the measured errors are attributable to minor localization differences rather than complete failures to identify roads. To further analyze model behaviour, the best-performing case, an average-performing case and the worst-performing case were selected based on their F1-score values. The best-performing sample was **case 93**, while the lowest-performing sample was **case 125**. A representative average-performing sample, **case 96**, was also selected since its metrics closely matched the global averages presented in Table 1. The quantitative results for these three cases are presented in Table 2.

Table 2: Comparison between selected test cases based on F1-score.

Metric	Best (case 93)	Avg. (case 96)	Worst (case 125)
Accuracy	99.35	98.89	98.49
Precision	71.87	63.69	49.98
Recall	80.71	66.67	41.97
F1-Score	76.03	64.98	45.62
IoU	61.33	48.51	29.55

Figures 2a, 2b and 2c respectively present Case 93, Case 96 and Case 125. In these visualizations, true positive road pixels are represented in green, false negatives in red and false positives in orange, while correctly identified background pixels are not shown.

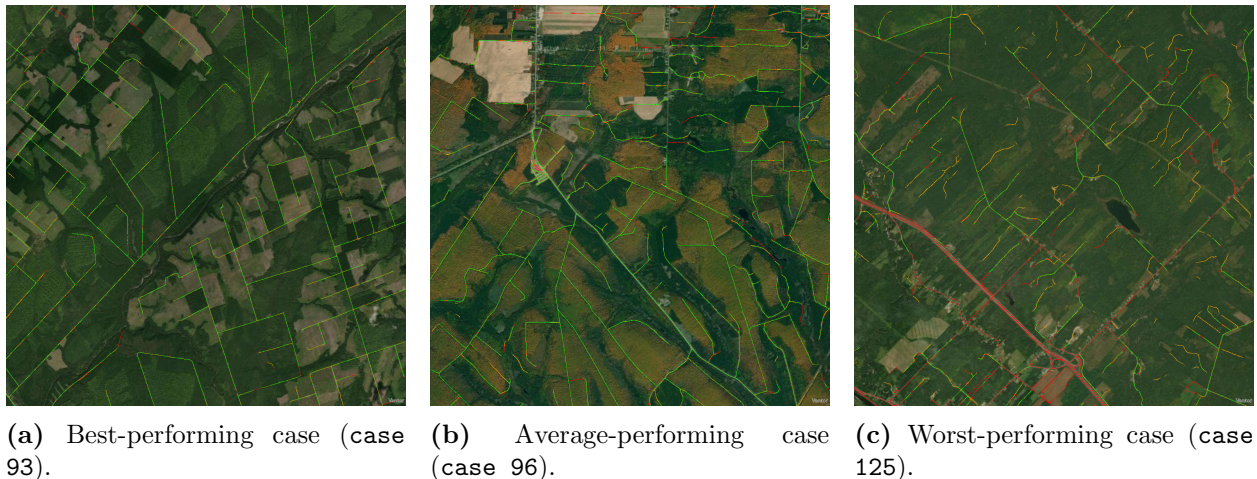


Figure 2: Examples of road segmentation performance for the best, average and worst-performing cases. True positive road pixels are represented in green, false negatives in red and false positives in orange.

Case 93 demonstrates the maximum performance achieved by the model. Most roads were correctly extracted with very limited false positive and false negative predictions. This behaviour is reflected by the F1-score of 76.03% and IoU of 61.33%. The model successfully identified the majority of agricultural and forestry roads while maintaining strong continuity along road segments and preserving thin linear structures. Only a few isolated prediction errors can be observed, generally occurring near road intersections, narrow road extremities or visually ambiguous regions.

Case 96 represents a typical or average-performing example. The metrics of this case closely match the global averages presented in Table 1, making it representative of the expected operational behaviour of the model. Most roads were successfully segmented with only moderate false positive and false negative predictions. The model was generally capable of following road network geometry while preserving road continuity and localization accuracy. Most prediction errors occurred along road boundaries or within short disconnected segments rather than through complete road omission. This behaviour suggests that the model can reliably extract the majority of forestry and agricultural roads even under moderately challenging conditions.

Case 125 represents the lowest-performing sample among the evaluated cases. In this example, a larger amount of false negatives and false positives can be observed, resulting in lower Precision, Recall, F1-score and IoU values. The primary source of error appears to be related to major paved roads and highways rather than forestry or agricultural roads. Visual inspection suggests that these errors are strongly influenced by annotation inconsis-

tencies, as paved roads were not consistently annotated as target roads within the dataset. Consequently, several correctly predicted paved roads were penalized as false positives, while unannotated paved road segments contributed to false negatives. Despite these inconsistencies, the model still performed relatively well on the intended forestry and agricultural road classes. This observation suggests that a portion of the measured errors likely originates from annotation quality and class-definition inconsistencies rather than purely from segmentation limitations.

Overall, the qualitative and quantitative analyses demonstrate that the proposed model consistently extracts forestry and agricultural roads across a wide range of environmental conditions. While performance naturally varies between cases, the model maintains relatively stable behaviour and produces useful road predictions suitable for large-scale operational deployment.

4.2 Threshold Sensitivity Analysis

To further analyze the behaviour of the proposed road segmentation model, a threshold sensitivity analysis was conducted using Case 96. The threshold is the value which a model will determine if a pixel is road or background, ranging from 0 to 1. Most often, this value is set at 0.5 and values above are positive, while below are negative. The objective of this experiment was to evaluate the impact of the prediction threshold on the balance between false positives and false negatives while also analyzing the evolution of common segmentation metrics such as Precision, Recall, F1-score and IoU. The experiment was performed using the final ONNX deployment model integrated within the ArcGIS Pro pipeline. Predictions were generated once using the probability output of the model and evaluated using thresholds ranging from 0.00 to 1.00 with increments of 0.05. The ground-truth raster was aligned to the prediction raster resolution and extent prior to evaluation to ensure correct pixel-level comparisons.

The results demonstrated that threshold selection has a relatively limited impact on overall segmentation performance within a broad intermediate range. Figure 3 shows the evolution of Precision, Recall, F1-score and IoU across all evaluated thresholds. At very low thresholds, particularly 0.00, nearly all pixels are classified as roads, resulting in extremely high Recall but very poor Precision and IoU. Conversely, at a threshold of 1.00, almost no road pixels are detected, causing Recall, F1-score and IoU to collapse toward zero. However, between approximately 0.10 and 0.90, segmentation performance remains remarkably stable. The best F1-score and IoU were obtained at a threshold of 0.40. At this threshold, the model achieved an F1-score of 69.87%, an IoU of 53.69%, a Precision of 67.88% and a Recall of 71.98%. The default threshold of 0.50 produced nearly identical performance, with an F1-score of 69.84% and an IoU of 53.66%. This negligible difference indicates that the proposed model is relatively robust to moderate threshold variations and does not require precise threshold tuning to maintain stable segmentation performance.

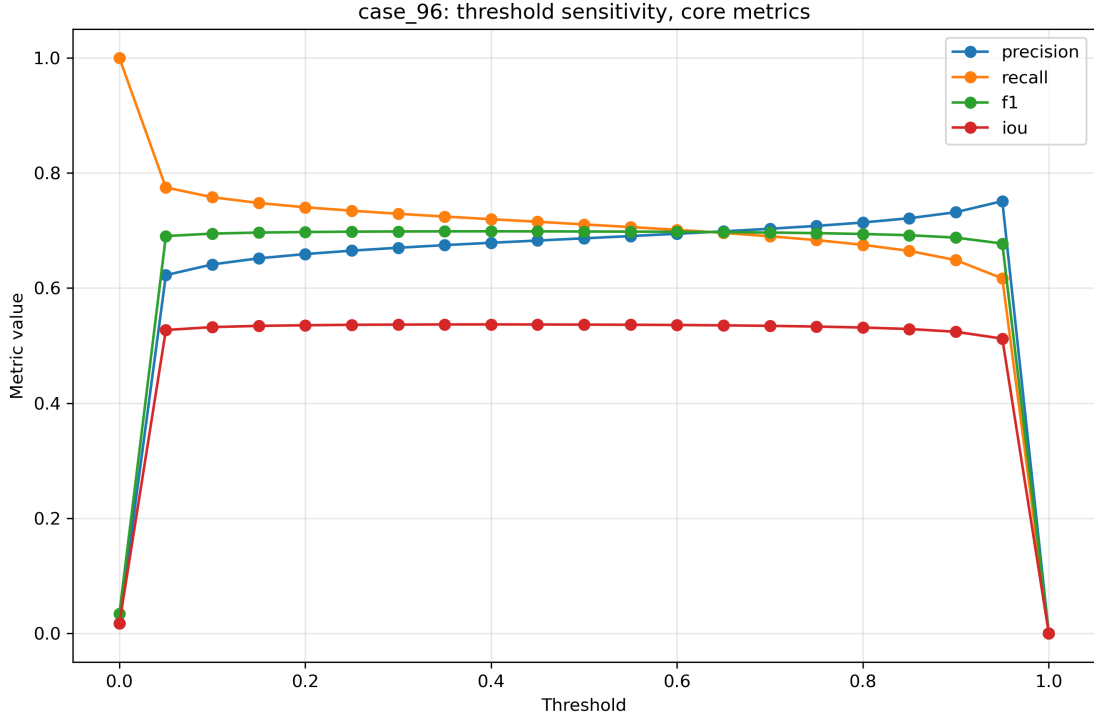


Figure 3: Evolution of Precision, Recall, F1-score and IoU for Case 96 across different threshold values.

The relationship between the False Positive Rate (FPR) and False Negative Rate (FNR) is illustrated in Figure 4. As expected, increasing the threshold gradually reduces the FPR while simultaneously increasing the FNR. At low thresholds, the model tends to over-segment roads, producing additional false positive detections. At higher thresholds, the model becomes more conservative, reducing false positives but missing more road segments. The FPR remained very low throughout most thresholds, generally below 1%, while the FNR increased progressively as the threshold increased. This behaviour is expected, as higher thresholds make the model more conservative, reducing false detections at the cost of missing additional road pixels.

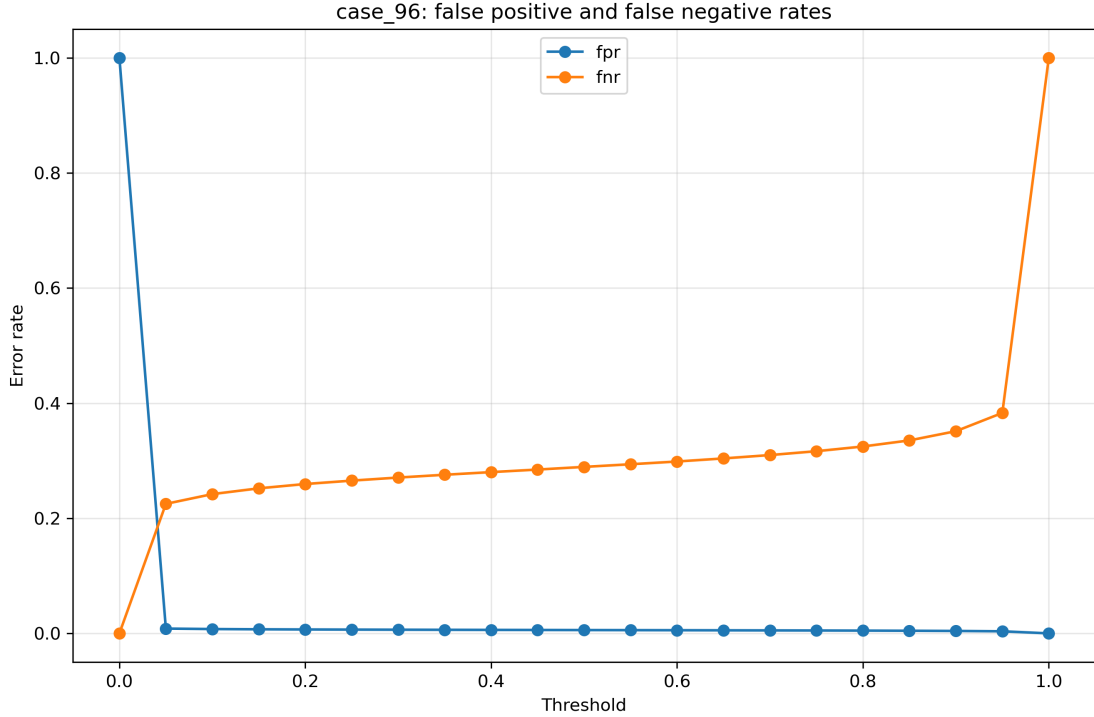


Figure 4: False Positive Rate and False Negative Rate for Case 96 across different threshold values.

Table 3 summarizes the quantitative results obtained for each evaluated threshold. The table further confirms that the model maintains relatively stable performance across a broad threshold range. The highest F1-score and IoU values were obtained at thresholds between approximately 0.35 and 0.50, while extreme threshold values significantly degraded segmentation performance. Very low thresholds resulted in excessive false positives and over-segmentation, whereas high thresholds progressively increased false negatives and under-segmentation.

Table 3: Threshold sensitivity results for Case 96.

Threshold	Precision (%)	Recall (%)	F1-score (%)	IoU (%)	FPR (%)	FNR (%)	Accuracy (%)
0.00	1.70	100.00	3.40	1.70	100.00	0.00	1.70
0.05	62.30	77.60	69.10	52.80	0.90	22.40	98.70
0.10	64.00	75.90	69.50	53.30	0.80	24.10	98.80
0.15	65.10	75.10	69.70	53.50	0.70	24.90	98.80
0.20	65.70	74.30	69.70	53.50	0.70	25.70	98.80
0.25	66.20	73.60	69.70	53.50	0.70	26.40	98.80
0.30	66.70	73.00	69.70	53.60	0.60	27.00	98.90
0.35	67.30	72.40	69.80	53.60	0.60	27.60	98.90
0.40	67.90	72.00	69.87	53.69	0.60	28.00	98.90
0.45	68.30	71.50	69.90	53.70	0.60	28.50	98.90
0.50	68.70	71.10	69.84	53.66	0.60	28.90	98.90
0.55	69.10	70.50	69.80	53.60	0.60	29.50	98.90
0.65	70.00	69.60	69.70	53.50	0.50	30.40	98.90
0.70	70.40	69.10	69.70	53.50	0.50	30.90	98.90
0.75	70.80	68.50	69.60	53.40	0.50	31.50	98.90
0.80	71.30	67.80	69.30	53.20	0.40	32.20	98.90
0.85	71.90	66.80	69.20	52.90	0.40	33.20	98.90
0.90	73.10	65.10	68.80	52.50	0.40	34.90	98.90
0.95	75.00	61.80	67.80	51.30	0.30	38.20	98.90
1.00	0.00	0.00	0.00	0.00	0.00	100.00	98.30

A visual confusion analysis for the best-performing threshold (0.40) is presented in Figure 5. True positives are shown in green, true negatives in black, false positives in orange and false negatives in red. The visualization demonstrates that the majority of road segments are correctly identified, while most errors occur along road boundaries, narrow road sections and partially ambiguous regions. False positives are generally sparse and localized, whereas false negatives mainly occur on weaker, visually unclear or partially occluded road segments. Most of the observed errors are therefore associated with localization and boundary delineation rather than complete failures to identify roads. Overall, the confusion visualization confirms the quantitative results and demonstrates the ability of the proposed model to maintain consistent segmentation quality throughout the evaluated region.

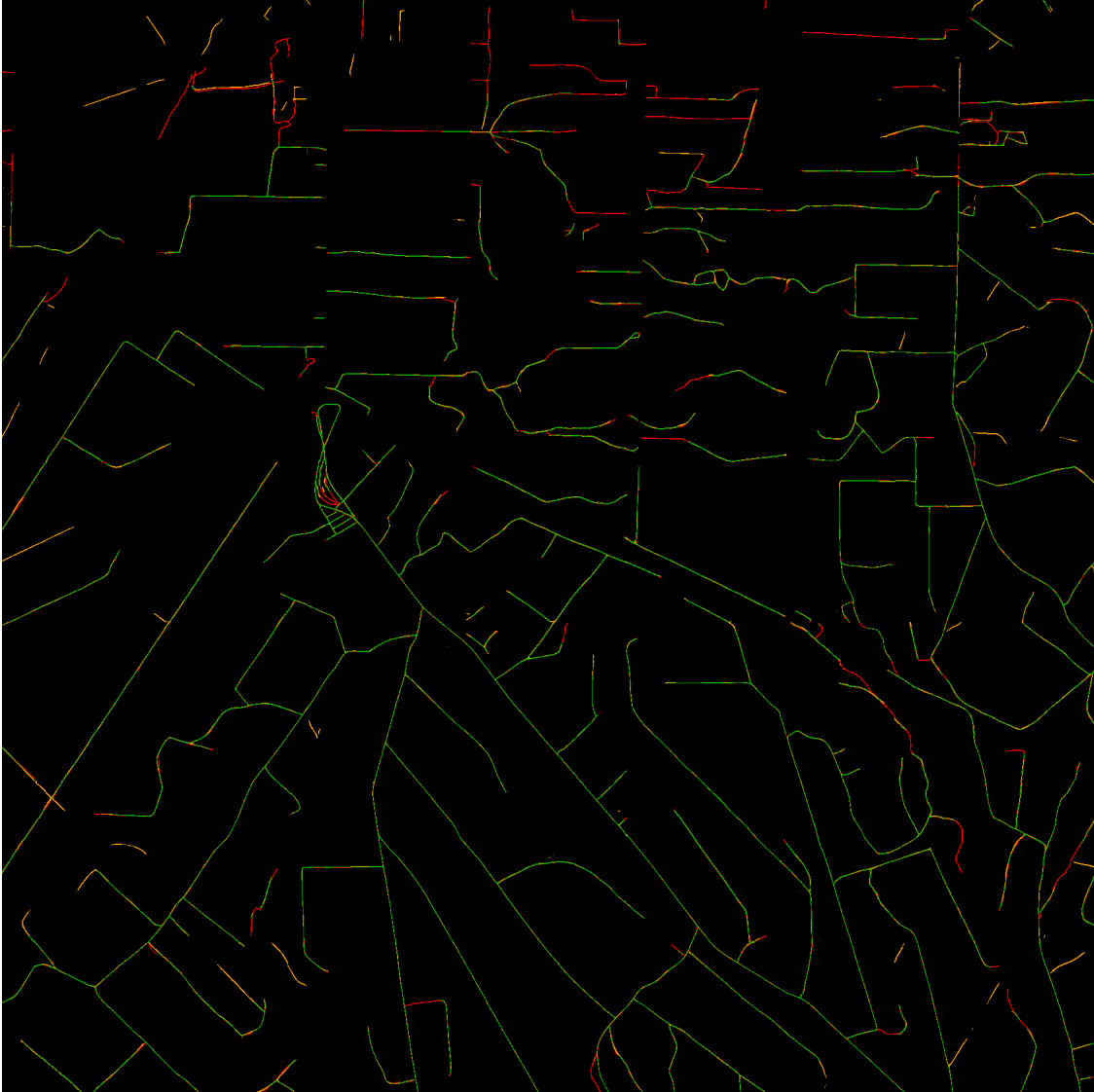


Figure 5: Confusion visualization for Case 96 at the best-performing threshold (0.40). Green corresponds to true positives, black to true negatives, orange to false positives and red to false negatives.

Overall, the threshold sensitivity analysis demonstrates that the proposed road segmentation model remains highly stable across a broad threshold range. The F1-score remains close to 70% between approximately 0.10 and 0.90, while the IoU remains near 53% throughout the same interval. Although the highest performance was obtained near a threshold of 0.40, the difference compared to the default threshold of 0.50 is negligible. Consequently, a threshold value of 0.50 was retained for deployment due to its balanced Precision–Recall trade-off, stable performance characteristics and compatibility with common segmentation practices.

Figure 6 presents the original RGB imagery and the corresponding probability distribution generated by the proposed segmentation model for Case 96. The left image shows the RGB

input provided to the model, while the right image presents the continuous probability map produced prior to thresholding. Rather than directly generating a binary road mask, the model first assigns a confidence value to each pixel representing the likelihood that the corresponding location belongs to a road. Dark blue regions correspond to probabilities close to 0, indicating strong confidence that the pixel belongs to the background class, while bright yellow regions correspond to probabilities close to 1, indicating strong confidence that the pixel belongs to a road. Intermediate purple and orange regions represent uncertain predictions and are generally observed near road boundaries, intersections or visually ambiguous regions. The probability map highlights the locations that the model considers most likely to correspond to roads and demonstrates that most road segments are assigned relatively high confidence values while most background regions remain close to zero. This behaviour explains why moderate threshold adjustments have only a limited impact on segmentation performance, as the majority of pixels are confidently classified and remain unchanged across a broad range of threshold values. The main issue in this sample is the complexity of the road network and overlapping colors between road and background pixels.

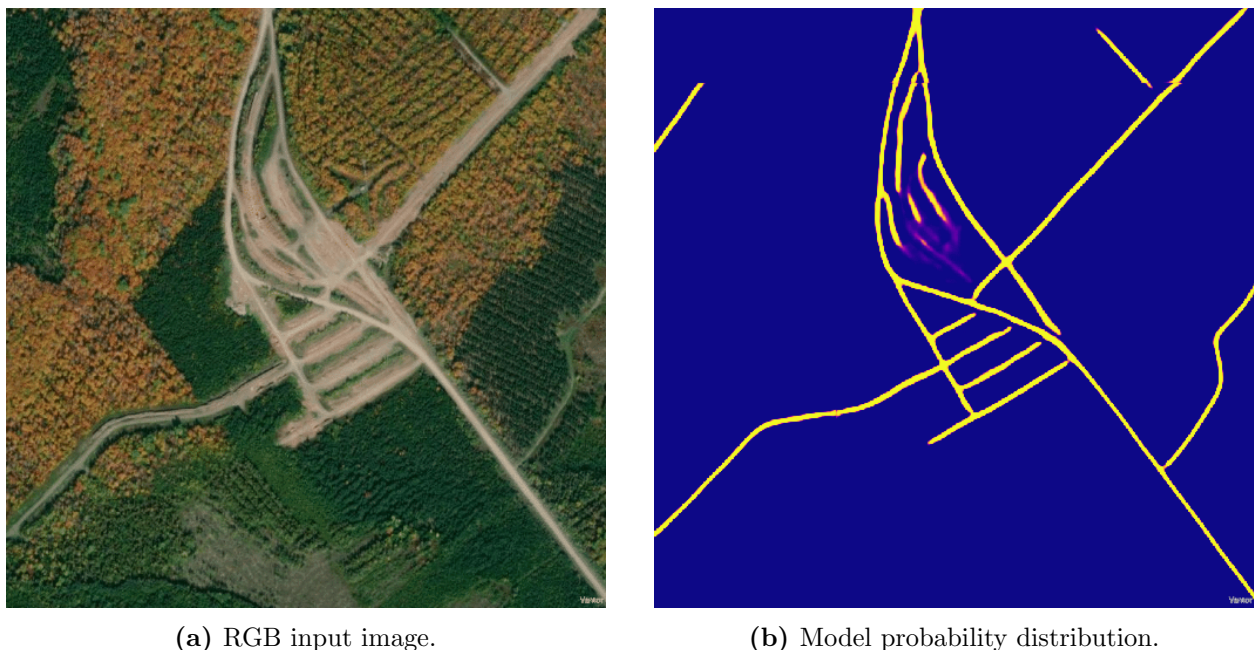


Figure 6: RGB imagery and corresponding probability distribution generated by the model for Case 96.

4.3 Model Efficiency and Deployment Performance

Model efficiency is an important consideration, particularly during deployment and operational integration, and heavily influenced the final model selection. Although larger architectures such as SegFormer and other transformer-based models could potentially achieve slightly higher segmentation performance, these approaches are substantially larger and require significantly more computational resources. Such models generally involve higher mem-

ory usage, slower inference times and more complex deployment pipelines, making them less suitable for rapid large-scale prediction and practical integration into lightweight GIS-based tools. As such, a U-Net architecture with a ResNet101 encoder was selected due to its relatively moderate size, stable performance and ease of implementation for further development and deployment. This architecture provides a balance between segmentation accuracy and computational efficiency while remaining compatible with consumer-grade hardware. The selected model contains approximately 59 million trainable parameters and over one hundred convolutional layers through the encoder-decoder structure, making it reasonably sized compared to transformer-based alternatives while still maintaining strong segmentation capabilities.

Performance experiments were conducted on consumer-grade hardware equipped with an NVIDIA RTX 3060 GPU, 32 GB of RAM and an Intel i9-10900 processor, representing a moderate desktop computer configuration accessible to many organizations and researchers. To evaluate deployment efficiency, Case 102 was used as a benchmark scenario. This case consisted of 49 image tiles covering an area of approximately 196 km². The complete deployment pipeline included four primary stages:

- tile export from ArcGIS
- ONNX prediction
- raster mosaicking
- clipping and post-processing

Using batch inference with a batch size of 8 and CUDA acceleration, the complete processing pipeline required only 48.86 seconds to process the full 196 km² study area. This corresponds to an average processing speed of approximately 1.00 second per tile and 0.25 seconds per km². The tile export stage required 13.40 seconds, representing approximately 27% of the total processing time. The ONNX prediction stage required 18.21 seconds, accounting for approximately 37% of the complete pipeline. However, a more detailed analysis reveals that only 5.86 seconds were spent performing actual neural network inference. The remaining prediction time consisted primarily of preprocessing operations (2.31 seconds) and prediction export and saving operations (9.74 seconds). This observation highlights an important characteristic of the deployment pipeline: the deep learning model itself is not the primary computational bottleneck. Pure ONNX inference represented only approximately 12% of the total pipeline execution time and achieved an average inference speed of 0.12 seconds per tile. In comparison, preprocessing, file generation and raster export operations collectively required more time than the neural network inference itself. The mosaicking stage required 5.50 seconds, while the final clipping and cleaning operations required 10.36 seconds. Together, these GIS processing operations accounted for approximately one-third of the total execution time. Consequently, future deployment optimizations are likely to benefit more from improvements to data handling, raster I/O and ArcGIS processing workflows than from additional model-level optimizations.

Table 4: Deployment efficiency results for Case 102 using ONNX inference and CUDA acceleration.

Metric	Value
Hardware	RTX 3060 / i9-10900 / 32 GB RAM
Case Area	196 km ²
Number of Tiles	49
Tile Size	(1024 × 1024)
Batch Size	8
Processing Times	
Export Time	13.40 s
Prediction Time	18.21 s
Preprocessing	2.31 s
ONNX Inference	5.86 s
Post-processing + Save	9.74 s
Mosaic Time	5.50 s
Clip + Clean Time	10.36 s
Total Pipeline Time	48.86 s
Average Processing Times	
Export per Tile	0.27 s
Prediction per Tile	0.37 s
Preprocessing per Tile	0.05 s
ONNX Inference per Tile	0.12 s
Post-processing per Tile	0.20 s
Mosaic per Tile	0.11 s
Total per Tile	1.00 s
Total per km ²	0.25 s
Estimated Processing Times	
CSRNO Coverage	(~30) min
New-Brunswick Coverage	(~5) h

Table 4 summarizes the deployment efficiency results obtained for Case 102. These results demonstrate that the proposed approach can efficiently process large geographic regions while maintaining practical deployment speeds on moderate consumer-grade hardware. The relatively low computational requirements also enable future integration into operational GIS workflows and potential near-real-time applications. By extrapolating these timing results, the proposed system could process the entire CSRNO region, which has an approximate area of 7,196.83 km², in roughly 30 minutes under similar hardware and deployment conditions. Similarly, extrapolation to the entire province of New-Brunswick, with an approximate area of 72,908 km², suggests that complete province-wide prediction could theoretically be achieved in approximately 5 hours using a single consumer-grade GPU system. Although these estimates provide a useful indication of the scalability of the proposed approach, they

are based solely on the total area covered and assume ideal processing conditions. In practice, actual processing times would likely be higher due to factors such as tile overlap, buffering operations, irregular region boundaries, image export overhead and additional GIS processing requirements. Furthermore, some deployments may require smaller strides or overlapping tiles to reduce edge artifacts and improve prediction continuity, increasing the total number of processed tiles. Consequently, a more realistic estimate would be approximately 1 hour for complete CSRNO coverage and approximately 10 hours for province-wide processing across New-Brunswick using the current deployment workflow and hardware configuration. Even under these more conservative estimates, the proposed approach remains highly practical for large-scale road extraction and environmental monitoring applications.

These results highlight the practical scalability of the proposed solution and demonstrate that large-scale forestry and agricultural road extraction can be performed efficiently without requiring high-end server infrastructure, cloud computing resources or specialized computational clusters. More heavily optimized implementations, particularly those avoiding GUI-based operations and relying exclusively on direct ONNX inference pipelines, could further improve deployment efficiency. Likewise, stronger hardware configurations equipped with more powerful GPUs, faster storage devices and increased memory bandwidth would further reduce processing times. Currently, one of the primary bottlenecks remains World Imagery extraction within ArcGIS. Unlike the neural network inference stage, which executes locally on the GPU, imagery must first be requested from remote ArcGIS servers during tile export operations. Consequently, network latency, server response times and raster export operations contribute substantially to the overall processing time. In practical deployments, storing imagery locally or processing pre-downloaded datasets would likely provide some of the largest performance improvements within the entire workflow.

4.4 Explainability

To provide additional insight into model behaviour, explainability analyses were conducted. As DL models are generally considered black-box approaches, explainability techniques can help better understand how convolutional neural networks generate predictions and which image regions contribute the most to the final output. Grad-CAM was used to generate explainability visualizations. This approach allows the selection of a convolutional layer within the network, typically one of the final encoder layers, to visualize the most activated regions associated with the prediction. By backpropagating gradients through the selected layer, Grad-CAM highlights the image pixels and structures that contributed the most to the final segmentation output. Explainability methods have shown strong benefits across multiple domains, particularly in applications where model interpretability and validation are important. In remote sensing applications, these techniques can help validate whether the model focuses on meaningful structures such as roads, edges and textures rather than unrelated background information. Figure 7 presents an example including the RGB image, the predicted segmentation mask and the corresponding Grad-CAM explainability visualization.

The results show that the model primarily focuses on linear road structures and surrounding road boundaries when generating predictions. Large portions of the road network are identified as highly important regions, while surrounding vegetation and background areas generally receive lower activation values. The visualizations also suggest that earlier extracted road-like structures are progressively refined through deeper convolutional layers to generate the final segmentation output. These observations indicate that the model learned meaningful spatial and contextual representations associated with forestry and agricultural roads.

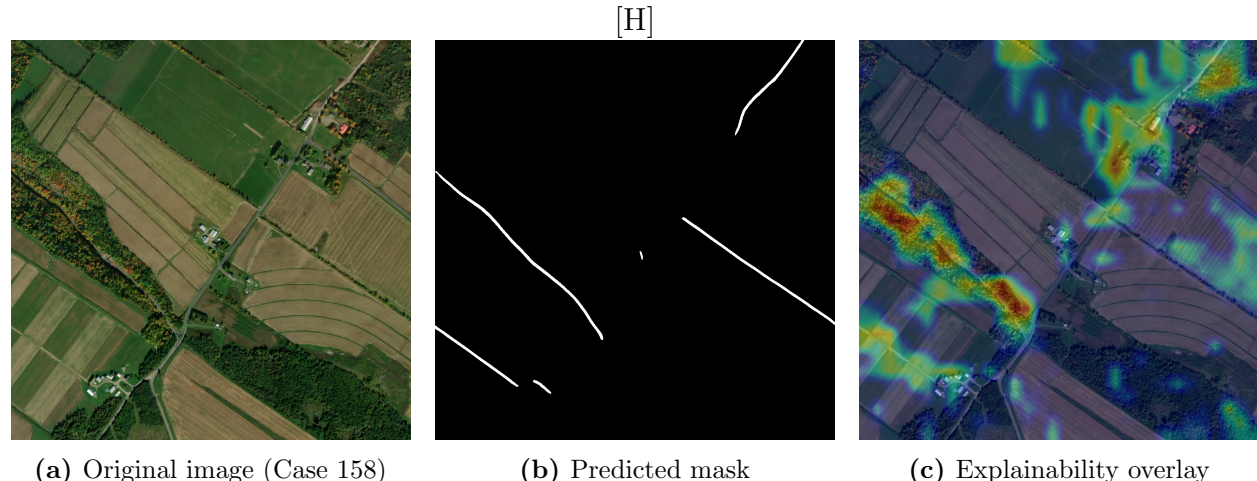


Figure 7: Example of road segmentation predictions and Grad-CAM explainability visualizations for three randomly selected samples.

5 Conclusion

The results obtained throughout this project demonstrate that deep learning provides an effective and practical solution for large-scale forestry and agricultural road extraction from high-resolution remote sensing imagery. The selected U-Net architecture with a ResNet101 encoder achieved strong segmentation performance while maintaining relatively low computational requirements and fast deployment speeds. Across all evaluated cases, the model successfully extracted the majority of forestry and agricultural roads while maintaining a very low false positive rate, making it suitable for operational mapping applications. The performance analysis demonstrated that the model achieved a good balance between precision and recall, allowing the extraction of road networks while limiting excessive false detections. Although some segmentation errors were observed, particularly in regions containing major paved roads, the analysis suggests that a portion of these errors originated from annotation inconsistencies rather than from limitations of the model itself. The Pixel Proximity Accuracy results further demonstrated that many apparent errors were caused by small spatial offsets between annotations and predictions rather than complete failures to detect roads. Overall, the qualitative and quantitative analyses indicate that the proposed

approach provides reliable road extraction performance across a wide range of environmental conditions.

The threshold sensitivity analysis further demonstrated the robustness of the model. Segmentation performance remained relatively stable across a broad range of threshold values, indicating that the model does not require extensive parameter tuning for deployment. Similarly, explainability analyses confirmed that the model learned meaningful road-related features and primarily focused on relevant linear structures when generating predictions. From an operational perspective, the deployment experiments demonstrated that the proposed solution is highly efficient. Using ONNX inference with CUDA acceleration, approximately 196 km² of imagery could be processed in less than one minute on a consumer-grade desktop computer. The results also showed that neural network inference itself represented only a small fraction of the total processing time, with most computational overhead originating from imagery extraction, raster generation and GIS processing operations. These findings highlight the practicality of the proposed solution for large-scale road mapping applications and demonstrate that province-scale deployment is achievable without requiring specialized computing infrastructure.

Overall, the developed model successfully meets the objectives of the project by providing an accurate, scalable and computationally efficient approach for automated road extraction. The combination of strong segmentation performance, rapid deployment capabilities and straightforward ArcGIS integration makes the proposed solution well suited for environmental monitoring, forestry management and road inventory applications throughout New-Brunswick and similar regions.

5.1 Future Directions and Next Steps

Several opportunities exist to further improve and expand the proposed system. The most immediate step consists of retraining the final model using the corrected annotations that were identified during the final validation phase. Although the impact of these annotation inconsistencies appears relatively minor, incorporating the corrected labels should improve dataset consistency and further increase model reliability. Future work could also focus on expanding the training dataset to include larger geographic regions and additional environmental conditions. While the current model was trained primarily using imagery from the CSRNO region, extending the dataset to other parts of New-Brunswick and the Maritime provinces would likely improve generalization and robustness. Similarly, the methodology could be adapted and fine-tuned for entirely new regions, provinces or countries with relatively limited effort due to the flexibility of the selected architecture.

Additional model optimization experiments could also be explored. Although the current model already provides a strong balance between performance and efficiency, future studies could investigate alternative architectures, hyperparameter optimization strategies, larger training datasets and additional data augmentation techniques. Given the relatively modest computational requirements of the selected architecture, such experiments can be performed

efficiently on consumer-grade hardware. From a deployment perspective, future improvements should primarily focus on workflow optimization rather than model optimization. The deployment analysis demonstrated that imagery extraction, raster generation and ArcGIS processing operations currently represent the largest bottlenecks within the pipeline. Consequently, local imagery storage, optimized raster handling procedures and reduced ArcGIS processing overhead could significantly decrease overall execution times. Further integration of direct ONNX processing workflows may also reduce deployment complexity and improve scalability.

Finally, the developed ArcGIS GUI provides a strong foundation for future development. Additional features such as automated batch processing, support for multiple road classes, integration with additional imagery sources, enhanced visualization tools and expanded reporting capabilities could further increase the usefulness of the system for operational users. Together, these improvements would support the development of a more versatile and scalable road extraction platform capable of supporting larger regions and additional environmental monitoring applications.